DATASHEET — INORGANIC EXPERIMENT 1 — WEEK 2

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$

Full name (as on student ID)	Matriculation no.	Date
KIM DONGHWAN	A0266094X.	2026/08/20 Thu.
Lab session ☐ Mon ☑ Thu	Fumehood no. B7.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
PPh_3 ✓	0.50 g	0.5076	
$RuCl_3 \cdot x\,H_2O$ ✓	0.08 g	0.080	
Dried absolute EtOH (to PPh_3)	20 mL	20	final
Dried absolute EtOH (to $RuCl_3$)	20 mL	20	final
Dried absolute EtOH (rinse)	5 mL	—	—
Cold dried EtOH (wash, 2 ×)	3 mL		

2. Reaction conditions

	Nominal	Actual	Remarks
Warming step 1: temp. / duration	~40 °C, 10–15 min		
Reflux start / end time			iphone timer / after it started to drip
Reflux duration	90 min		~110°C.
Oil bath temp. / hotplate setting	< 130 °C	110 – 120 ↓	setting 200°C / 300 rpm.
Vacuum drying duration	20–30 min	26 mins.	

3. Product

	Value	Remarks
Mass of empty vial / paper	1.153 g.	
Mass of vial + dried product	1.395	
Mass of $RuCl_2(PPh_3)_3$	0.242 g.	
Colour of product	brown dark	redbrown - like color?
Mass submitted for ESI-MS		
IR method ☐ ATR ☐ KBr		

4. Observations

Record colour changes with the time at which they occurred: the two starting solutions, on mixing, during reflux, on cooling, of the filtrate and of the washings.

To be completed at the end of the session		
Student signature	GTA name	GTA signature 20/08/2026

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

DATASHEET — INORGANIC EXPERIMENT 1 — WEEK 3

Session II: Control Study — Oxidation *without* Catalyst

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
Dried acetone	8 mL		
1-Phenylethanol	0.06 mL		
K_2CO_3	0.07 g	0.0704	
n-Hexane (quench / rinse 2×)	50 / 5 mL		
Mesitylene (internal standard)	0.07 mL		
$CDCl_3$	0.6 mL		

2. Reaction conditions

	Nominal	Actual	Remarks
Reflux start / end time	—		
Reflux duration	60 min		
Oil bath temp. / hotplate setting	< 100 °C		
Rotary evaporator bath temp. / pressure	—		
Vacuum drying of oil	15–20 min		

3. Conditions to reproduce in Session III

Write these down now. Next week's comparison depends on matching them.

Solvent batch / bottle	Ambient conditions

4. Observations

Colour on charging, during reflux and after cooling; appearance of the Celite pad and of the filtrate; appearance of the oily residue after concentration.

_____

_____

_____

5. NMR sample

☐ Tube labelled with name, session and S/N no. ☐ Submitted S/N no. _____

To be completed at the end of the session		
Student signature	GTA name	GTA signature Song Tan 2/08/2026

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DATASHEET — INORGANIC EXPERIMENT 1 — WEEK 4

Session III: Catalytic Oxidation *with* $RuCl_2(PPh_3)_3$

Full name (as on student ID)	Matriculation no.	Date
KIM. Dong Hwan	A0266044X.	
Lab session ☐ Mon ☑ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
Dried acetone (fixed)	8 mL ✓		
1-Phenylethanol (fixed)	0.06 mL ✓		
K_2CO_3	0.07 g ✓	0.070	
$RuCl_2(PPh_3)_3$ (Week 2 product)	0.06 g	0.060	
n-Hexane (quench / rinse 2×)	50 / 5 mL		
Mesitylene (internal standard) (fixed)	0.07 mL		
$CDCl_3$	0.6 mL		

2. Reaction conditions

	Nominal	Actual	Remarks
Reflux start / end time (phone time)	— ✓		
Reflux duration	60 min		
Oil bath temp. / hotplate setting	< 100 °C		
Vacuum drying of oil	15–20 min	15 min (timed by phone)	

3. Colour log — record the time with each entry

On adding the catalyst		Celite pad after filtration	
During reflux		First filtrate	
After cooling		Residue after rotary evaporation	

Second filtration required? ☐ No ☐ Yes — reason: _____

add pattern in thin.

4. Deviations from the Session II conditions

Solvent batch, ambient conditions, reflux duration, anything not matched

5. Other observations

· greenish color,

6. NMR sample

☐ Tube labelled with name, session and S/N no. ☐ Submitted S/N no. _____

To be completed at the end of the session		
Student signature	GTA name	GTA signature: Sang Jun 3/9/2026

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

2026/3/9/Thu.